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Lab 09 - Probability & Statistics

R Script

```r
# Part i

baking_times <- rnorm(25, mean = 45, sd = 2)
print(baking_times)

# Part ii
test_result <- t.test(baking_times, mu = 46, alternative = "less")

print(test_result)
```

Output

```
> # Exercise 1
> # Part i
>
> baking_times <- rnorm(25, mean = 45, sd = 2)
> print(baking_times)
 [1] 43.77935 48.14682 40.33206 42.53151 39.38466 46.24175 40.18897 46.62225 43.85320 46.88741 41.72602 42.04457 44.61694 42.86783
[15] 46.58906 43.57522 45.89430 46.44392 46.72526 42.53279 41.04428 48.42615 46.42554 45.50741 48.29735
>

> # Part ii
> test_result <- t.test(baking_times, mu = 46, alternative = "less")
>
> print(test_result)

        One Sample t-test

data:  baking_times
t = -2.9036, df = 24, p-value = 0.003897
alternative hypothesis: true mean is less than 46
95 percent confidence interval:
     -Inf 45.35402
sample estimates:
mean of x
 44.42738
```